



**CALICUT UNIVERSITY – FOUR-YEAR
UNDERGRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	DIGITAL ELECTRONICS				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 – 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY2CJ101- ELECTRONICS I & PHY6CJ305- ELECTRONICS II				
Course Summary	The course covers the design and analysis of combinational logic circuits, sequential circuits using flip-flops, counters, and registers, as well as techniques for interfacing digital systems with the analog world, providing a comprehensive understanding of digital logic design principles and applications.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Design and analyze combinational logic circuits	Apply	Procedural Knowledge	Homework assignments, exams

CO2	Implement sequential circuits using flip-flops	Apply	Procedural Knowledge	Laboratory experiments, projects
CO3	Design and construct various types of counters	Create	Procedural Knowledge	Design projects, simulations
CO4	Analyze the operation of registers in digital systems	Understand	Conceptual Knowledge	Quizzes, concept tests
CO5	Interface digital systems with analog components	Apply	Procedural Knowledge	Case studies, practical exams
CO6	Evaluate and troubleshoot digital-analog interfaces	Analyze	Procedural Knowledge	Laboratory reports, demonstrations
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	COMBINATIONAL LOGIC CIRCUITS		10	16
	1	Sum-of-Products Form, Simplifying Logic Circuits, Algebraic Simplification	2	
	2	Designing Combinational Logic Circuits, Karnaugh Map Method, Exclusive-OR and Exclusive-NOR Circuits	4	
	3	Parity Generator and Checker	2	
	4	Enable/Disable Circuits	2	
	Sections 4.1-4.8 of Book 1			

II	FLIP-FLOPS AND RELATED DEVICES		12	18
	5	Clocked S-R Flip-Flop, Clocked J-K Flip-Flop, Clocked D Flip-Flop, D Latch (Transparent Latch)	3	
	6	Asynchronous Inputs, Flip-Flop Timing Considerations, Potential Timing Problem in FF Circuits	2	
	7	Flip-Flop Applications, Flip-Flop Synchronization, Detecting an Input, Sequence, Detecting a Transition or “Event”	2	
	8	Data Storage and Transfer, Serial Data Transfer: Shift Registers, Frequency Division and Counting, Application of Flip-Flops with Timing Constraints	3	
	9	Microcomputer Application, Schmitt-Trigger Devices, One-Shot (Monostable Multivibrator)	2	
	Sections 5.6-5.23 of Book 1			
III	COUNTERS AND REGISTERS		12	18
	10	Asynchronous (Ripple) Counters, Propagation Delay in Ripple Counters, Synchronous (Parallel) Counters,	3	
	11	Counters with MOD Numbers $< 2N$, Synchronous Down and Up/Down Counters	3	
	12	Presetable Counters, IC Synchronous Counters	2	
	13	Register Data Transfer, IC Registers	2	
	14	Shift-Register Counters	2	
	Sections 7.1-7.7, 7.15-7.17 of Book 1			
IV	INTERFACING WITH THE ANALOG WORLD		14	18
	15	Review of Digital Versus Analog, Digital-to-Analog Conversion	1	
	16	DAC Circuitry, DAC Specifications	2	

	17	An Integrated-Circuit DAC, DAC Applications	2	
	18	Analog-to-Digital Conversion	2	
	19	Digital-Ramp ADC, Data Acquisition	1	
	20	Successive-Approximation ADC, Flash ADCs	2	
	21	Sample-and-Hold Circuits, Multiplexing	2	
	22	Digital Signal Processing (DSP), Applications of Analog Interfacing	2	
	Sections 11.1-11.6,11.8-11.12,11.15-11.18 of Book 1			
V	OPEN ENDED MODULE- MEMORY DEVICES		12	
	Books and References: 1. Digital systems principles and applications by Moss, Gregory L._ Tocci, Ronald J._ Widmer, Neal S -Pearson, 12 Edition (Book 1) 2. Digital Design" by M. Morris Mano and Michael D. Ciletti 3. Digital Electronics: Principles and Applications" by Roger L. Tokheim 4. Digital Fundamentals" by Thomas L. Floyd and David M. Buchla 5. Digital Logic and Computer Design" by M. Morris Mano			

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PS O4	PS O5	PS O6	PO1	PO2	PO3	PO4	PO5	PO 6	PO 7
CO 1	2	0	2	3	3	2	2	2	0	0	2	2	1
CO 2	2	2	2	3	3	2	2	2	0	0	2	2	1
CO 3	2	2	3	3	3	2	2	2	0	0	2	2	1
CO 4	1	2	2	3	3	2	2	2	0	0	2	2	1
CO 5	2	2	2	2	3	2	2	2	0	0	2	2	1
CO 6	2	2	3	3	3	2	2	2	0	0	2	2	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory/ Practical Exam	Assignment /Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	